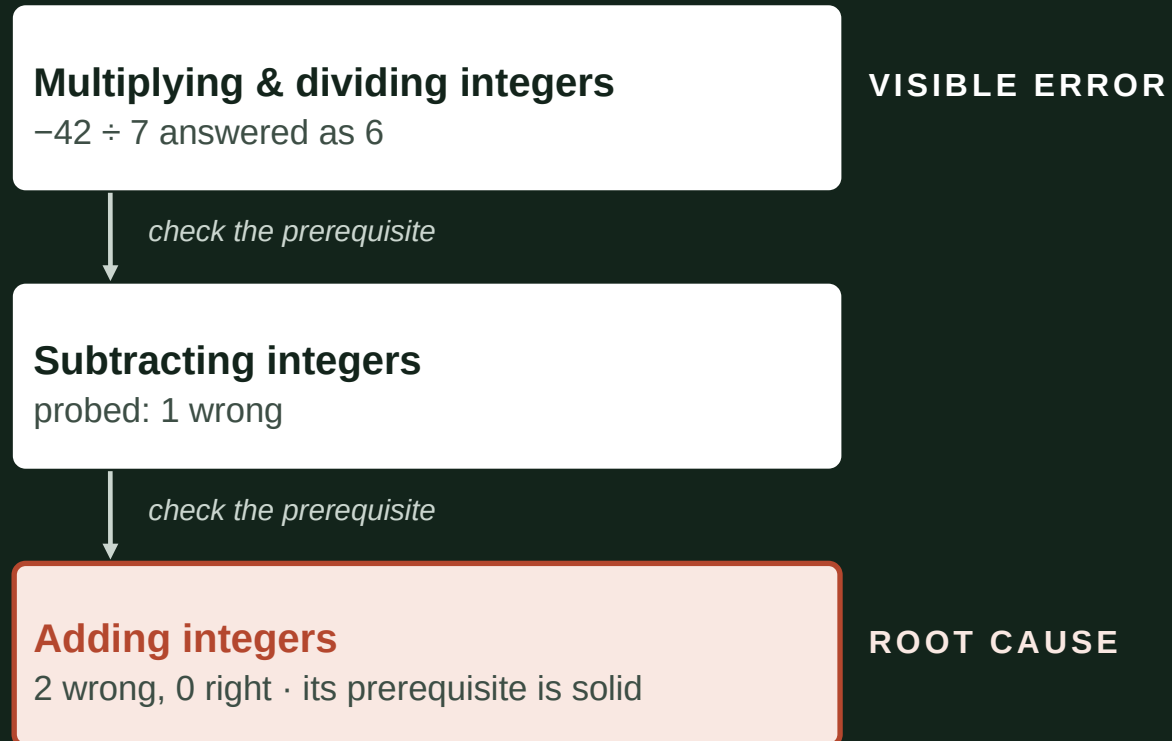


Visible error $\neq$ root cause

EXAMPLE FROM THE DEMO CONTENT



WHAT JUTHOOR DOES

- 1 Evidence first**
No root from one mistake: errors are counted across the chain
- 2 Prerequisite-aware**
Investigates earlier lessons along a prerequisite graph
- 3 Honest about uncertainty**
Says when the evidence is not enough yet, and what would settle it
- 4 Closes the loop**
Practice on the root, retry of the original lesson, mastery update

B2C: the learner practises; the parent (the buyer) sees where the gap started and what happens next.

One complete loop, on real content

Objective

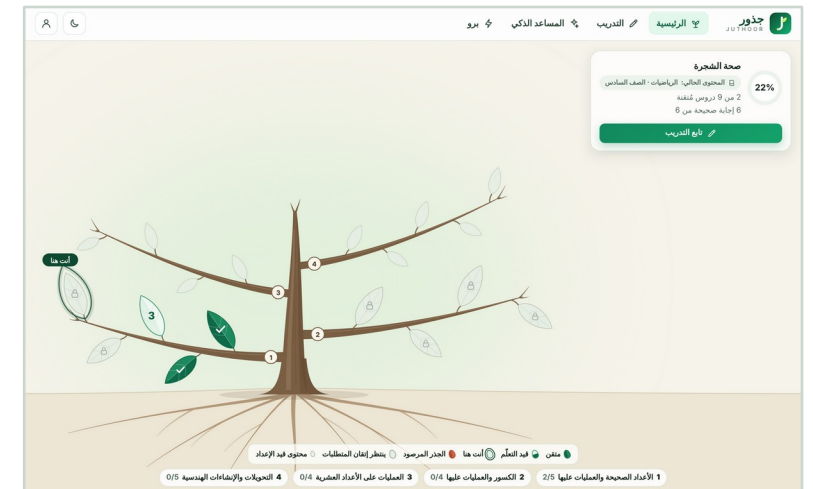
Show that one learner's wrong answers can be traced to a most-likely prerequisite gap, with evidence and confidence, and that the system then remediates it and returns the learner to the original lesson.

IN THE MVP

- Full loop: evidence → root → remediation → retry → resolved
- 9 live lessons, 140 question templates (demo content: Grade 6 maths)
- BKT mastery and rule-based diagnosis with a confidence level
- Learner diagnosis card and parent report
- Accounts, roles, ownership checks, Free and Pro plans
- Optional AI tutor with an offline fallback

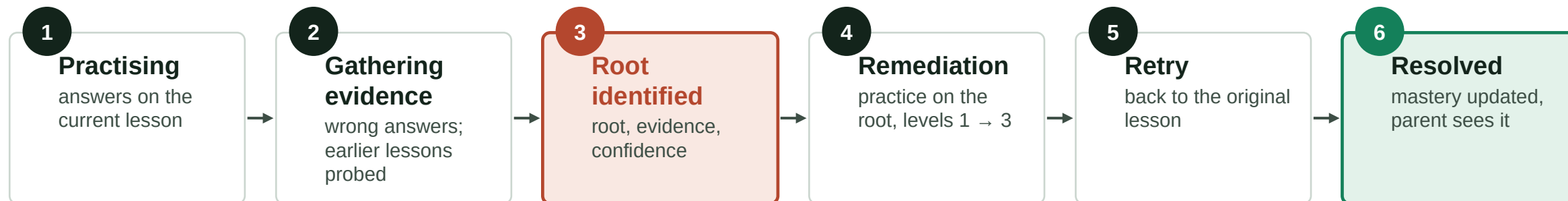
NOT IN THE MVP

- Units 3–4 and other subjects
- Field results from real learners
- Calibrated model parameters
- Production deployment, load testing
- Teacher or school product (B2C by design)
- Community features (hidden in judge mode)



The learner's curriculum tree: current lesson marked “you are here”, 9 of 18 lessons live.

Evidence first, then a root, then recovery



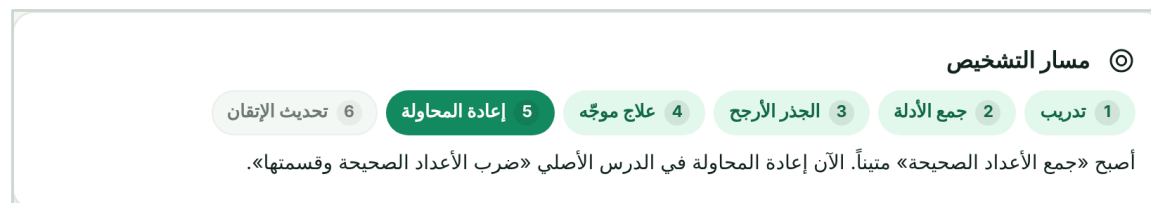
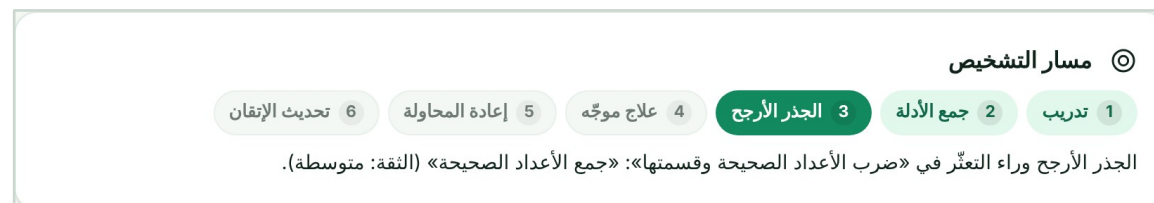
Not enough evidence yet

The learner sees why (for example: 1 error on the chain, 3 needed) and what would settle it. If one more answer can decide, the engine asks a confirmation question.

A root is named only if

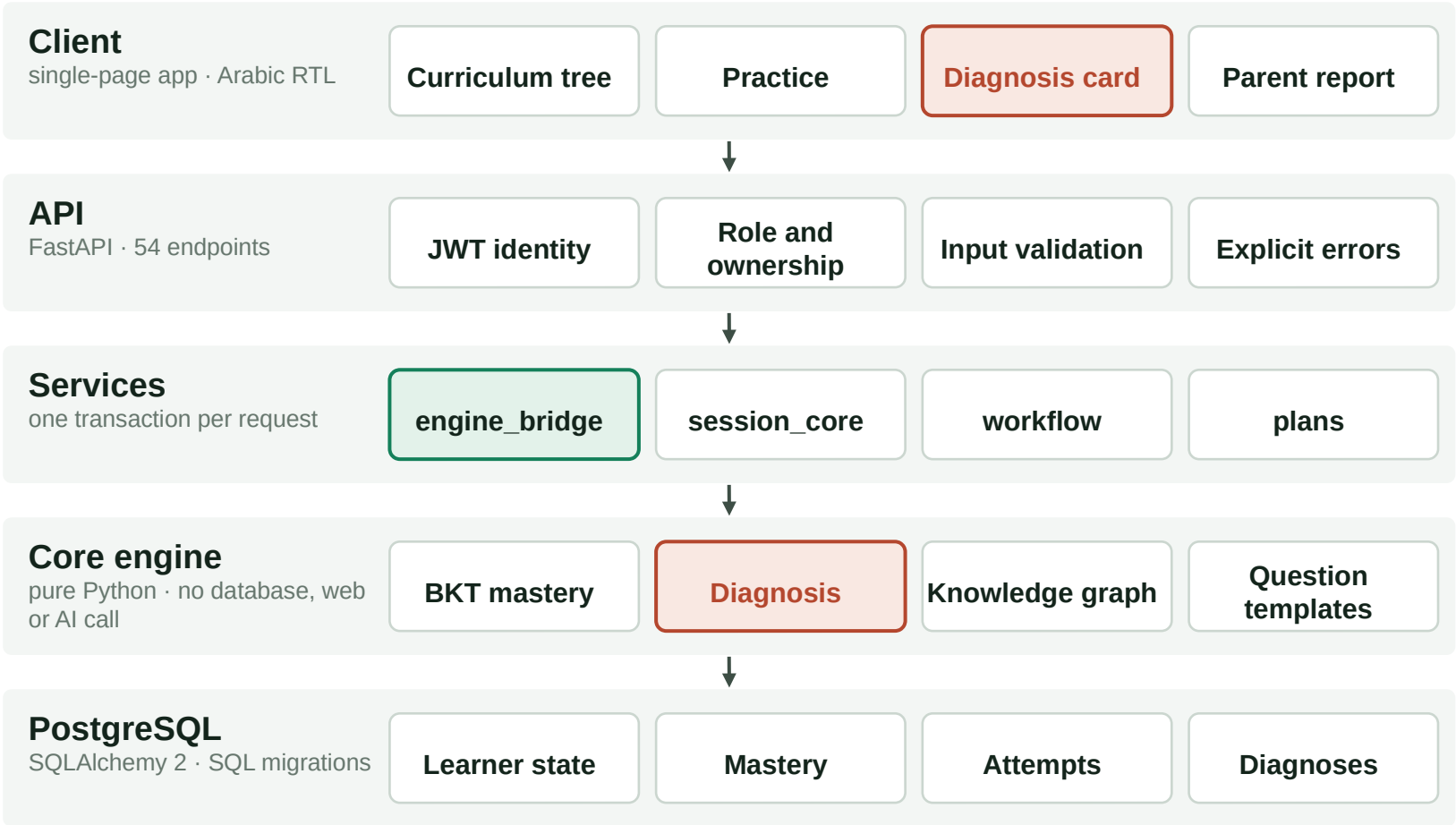
≥ 3 wrong answers on the chain · ≥ 2 on the root · likelihood ratio ≥ 20 · its prerequisites are solid. Confidence is a level (High / Medium), not a probability.

THE SAME STAGES IN THE APP (ARABIC, RIGHT-TO-LEFT, COMPUTED ON THE SERVER)



Core diagnosis runs without any AI service

CORE REQUEST PATH



SUPPORTING · OPTIONAL

- AI tutor**
OpenAI or Groq · 6 s timeout, offline fallback
- Vector retrieval**
ChromaDB · off unless enabled
- Stripe Checkout**
else a labelled demo checkout
- SMTP**
password reset codes

Core diagnosis needs no AI
Grading and diagnosis run in the pure engine and call none of the services above. If a provider is absent or fails, the core loop runs unchanged.

From wrong answers to a named root, and back

1 · VISIBLE ERROR



Happened. $-42 \div 7$ answered as 6.

Observed. 1 error on the chain; 3 needed.

Decided. No verdict yet. Probe the prerequisite.

2 · ROOT NAMED



Observed. Adding integers: 2 wrong, 0 right; its prerequisite is solid.

Decided. Root: adding integers. Confidence Medium.

Why. Thresholds met; the gap is not further down.

3 · RECOVERY AND RETURN



Action. Practice on the root, then retry the original lesson.

After. Root mastery 23 % → 98 %; stage resolved.

EVIDENCE THE ENGINE USED (DEMO PATH)

Lesson	Wrong	Right	Verdict
Multiplying & dividing (visible)	4	0	explained by prerequisite
Subtracting integers	1	0	explained by prerequisite
Adding integers	2	0	root · Medium confidence
Comparing integers	0	3	solid

Screens: the MVP frontend on the E2E test harness with the real engine.

7

wrong answers before a root was named

23 % → 98 %

live mastery of the root during recovery

6 / 6

stages reached, through to resolved

Simulated learners (810, deterministic)

Clean answers: right root in 97.1 % of the cases where one was named; 0 % premature. Not accuracy on real students.

Tested in layers, limits stated plainly

PYTEST · 382 TEST FUNCTIONS · 829 CASES

129

engine, graph, logic
no database

122

API + PostgreSQL
scenarios A–S,
retries, outages

37

security and B2C
ownership, Child ID,
plans

50

tutor and safety
guardrail, offline
fallback

Full pytest, PostgreSQL, final revision: 829 passed, 0 failed

Team machine, as reported. Preflight (44 live checks): not yet run on this revision. Plus 44 tests on config, colours, finance and migrations.

OTHER CHECKS · RE-RUN 5 OCT 2026 HERE

223

UI checks
223/223 passed

26

real-browser checks
26/26 passed

4

SQL scripts
PostgreSQL 16, all
pass

810

simulated learners
benchmark, same
output every run

Current limitations

- One content pack: 9 of 18 lessons live, one subject
- No field results yet; no real-student accuracy claim
- BKT parameters and thresholds are uncalibrated defaults
- Prerequisite edges not yet reviewed by independent experts
- Single-worker rate limiting; no HTTP load test
- Tutor answers cannot be guaranteed correct

Backup slides for questions

The six slides before this one are the judging presentation. The slides after it are not part of the main flow; they support technical questions.

B1

Diagnosis rules

when a root is named, when the engine abstains, how confidence is set

B2

Integration and failure handling

one answer end to end; what each error response means

B3

Simulated benchmark

810 simulated learners, labelled as simulation only

When a root is named, and when it is not

A ROOT IS NAMED ONLY IF

- ≥ 3 wrong answers on the chain
- ≥ 2 wrong answers on the root
- Likelihood ratio ≥ 20 against “no gap”
- Every direct prerequisite is solid (≥ 2 right, 0 wrong, or mastered)

Each wrong answer multiplies the ratio by 8, each right answer by 0.22 (P(correct | gap) = 0.2, P(correct | known) = 0.9).

OTHERWISE: INSUFFICIENT_EVIDENCE

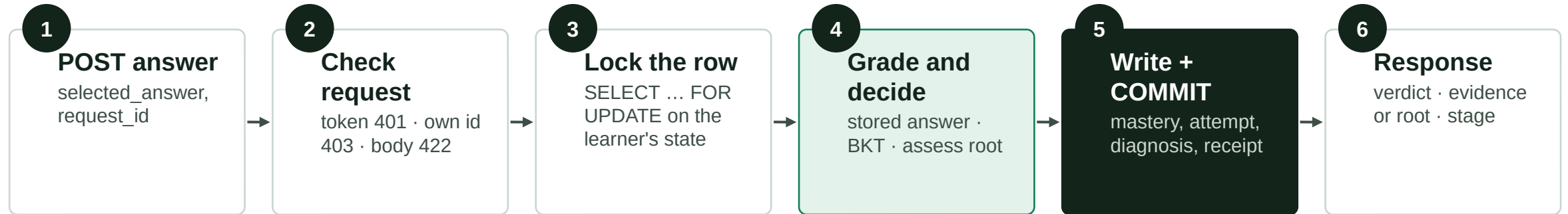
Reason	What would settle it
too_few_errors	More answers on the chain
root_needs_confirmation	One confirmation probe
prerequisites_unverified	Answers on that prerequisite
mixed_evidence	More answers to resolve the mix

CONFIDENCE LEVELS (A RULE)

- High**
≥ 3 wrong on the root, ≥ 75 % wrong, prerequisites observed correct
- Medium**
≥ 2 wrong on the root and more wrong than right; the cap when a rival root exists
- Low**
not reachable with the default thresholds

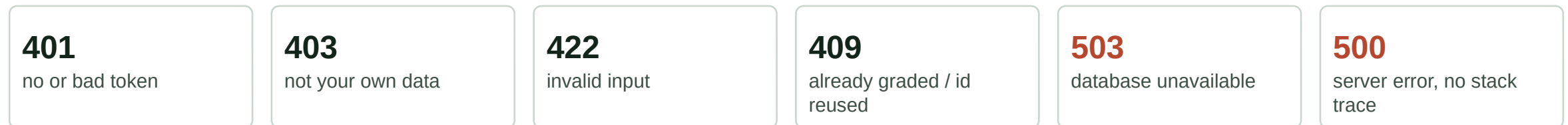
Source: app/engine/diagnosis.py · tested in tests/test_diagnosis_engine.py (incl. diamond and reading-literacy graphs)

One answer, end to end



Steps 3–5 run in one database transaction; any error rolls everything back.

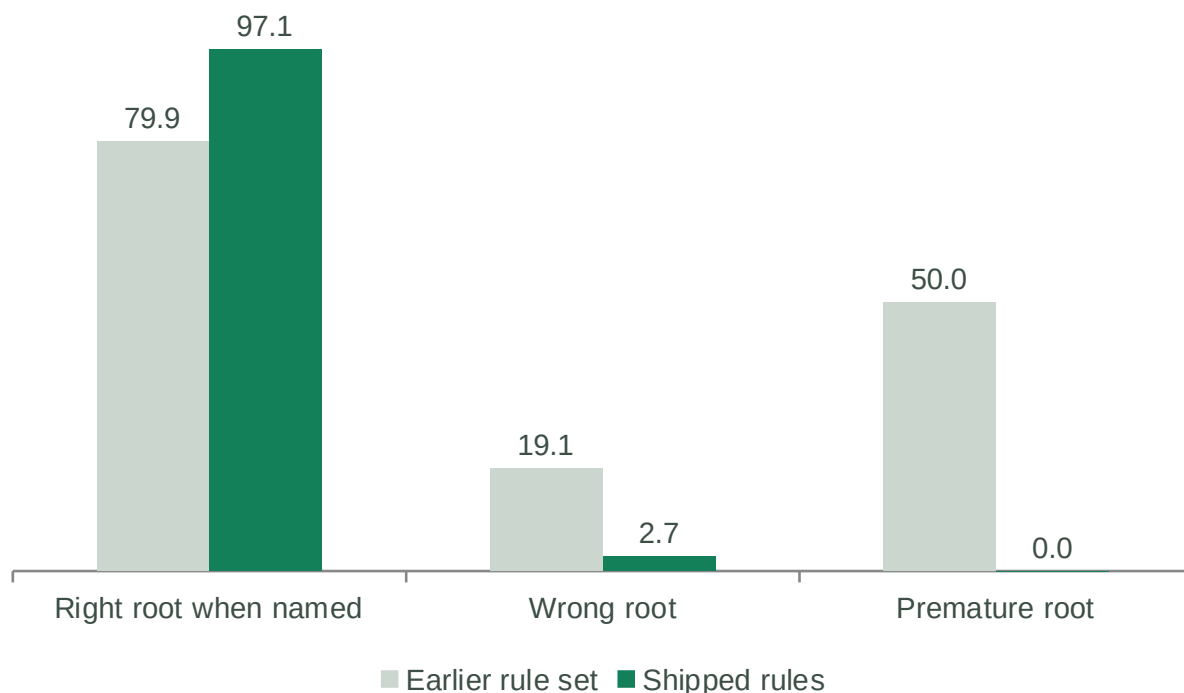
RESPONSES WHEN SOMETHING IS WRONG



A retry with the same request_id returns the stored response instead of grading twice. The AI tutor fails over to an offline tutor after a 6 s timeout; diagnosis never depends on it.

Measured on simulated learners only

Known root, clean answers (%), 225 cases



Shipped rules	Right when named	Wrong root	Abstained
Clean answers	97.1 %	2.7 %	7.6 %
Noisy answers	91.1 %	7.6 %	14.7 %
Lucky guesses	62.4 %	30.2 %	19.6 %

Learners with no gap: false diagnosis 2.2 % (clean), 8.9 % (noisy), 20.0 % (very noisy).

How to read this

810 simulated cases, fixed seeds, same output every run. It measures the rules, not real learners. Overall right with clean answers: 89.8 %, because some cases correctly abstain. Learners who guess right remain a weak case.

Source: scripts/diagnostic_benchmark.py --reps 5 · benchmarks/before.json · docs/COMPETITIVE_ADVANTAGE.md